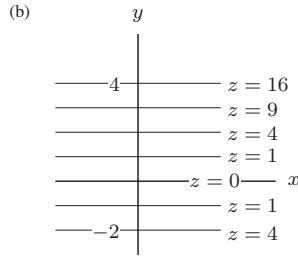
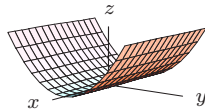
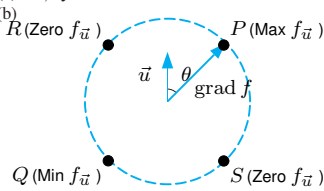
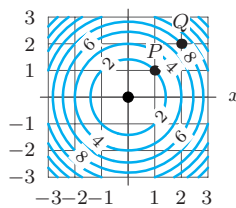


- 9 $\sin \theta \vec{i} + r \cos \theta \vec{j}$
 11 $\nabla z = \frac{1}{y} \cos\left(\frac{x}{y}\right) \vec{i} - \frac{x}{y^2} \cos\left(\frac{x}{y}\right) \vec{j}$
 13 $\left(\frac{-12\beta}{(2\alpha-3\beta)^2}\right) \vec{i} + \left(\frac{12\alpha}{(2\alpha-3\beta)^2}\right) \vec{j}$
 15 $60\vec{i} + 85\vec{j}$
 17 $10\pi\vec{i} + 4\pi\vec{j}$
 19 $(\pi/2)^{1/2} \vec{i}$
 21 $\frac{1}{100}(2\vec{i} - 6\vec{j})$
 23 $-46/5$
 25 $22/5$
 27 $84/5$
 29 $(2x + 3e^y)dx + 3xe^y dy$
 31 $(x+1)ye^x \vec{i} + xe^x \vec{j}$
 33 Negative
 35 Positive
 37 Positive
 39 $-\vec{j}$
 41 $\vec{j}$
 43 $-\vec{i} - \vec{j}$
 45 $\vec{i} - \vec{j}$
 47 (a) (3.96838, 5.09487)
 (b) 0.1052
 (c) $1/(3\sqrt{10})$
 49 4.4
 51 (a) $-16\vec{i} + 12\vec{j}$
 (b) $16\vec{i} - 12\vec{j}$
 (c) $12\vec{i} + 16\vec{j}$; answers may vary
 53 (a) -3.268
 (b) -4.919
 55 (a) Should be number
 (b) 11/5
 57 -0.5; better estimate is -1.35
 59 -1.1; better estimate is -1.8
 61 $4\vec{i} + 6\vec{j}$; $4(x-2) + 6(y-3) = 0$
 63 $-4\vec{i} + \vec{j}$; $-4(x-2) + (y-3) = 0$
 65 (a) Negative
 (b) Negative
 67 (a) Negative
 (b) Positive
 (c) Positive
 (d) Negative
 69 Yes
 71 Yes
 73 (a)



- (c) $\vec{j}$
 75 (a) Circles centered at P
 (b) away from P
 (c) 1
 77 $(3\sqrt{5} - 2\sqrt{2})\vec{i} + (4\sqrt{2} - 3\sqrt{5})\vec{j}$
 79 $\frac{4\sqrt{2}}{6\vec{i} + 2\vec{j}}$
 81 $5/\sqrt{2}$
 83 (a) P, Q
 (b) 
 (c) $\|\text{grad } f\|$
 $f_u = \|\text{grad } f\| \cos \theta$
 89 $f_u(0, 0)$ is scalar, not vector
 91 Closer contours give longer gradients
 93

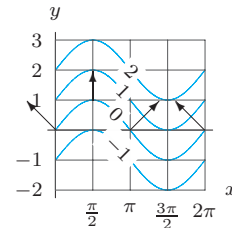


- 95 False
 97 False
 99 True
 101 False
 103 True
 105 True
 107 True
 109 True
 111 False

Section 14.5

- 1 $2x\vec{i}$
 3 $e^x e^y e^z (\vec{i} + \vec{j} + \vec{k})$
 5 $\frac{-2xyz^2}{(1+x^2)^2} \vec{i} + \frac{z^2}{1+x^2} \vec{j} + \frac{2yz}{1+x^2} \vec{k}$
 7 $(x\vec{i} + y\vec{j} + z\vec{k}) / \sqrt{x^2 + y^2 + z^2}$
 9 $y\vec{i} + x\vec{j} + e^z \cos(e^z) \vec{k}$

- 11 $e^p \vec{i} + (1/q)\vec{j} + 2re^{r^2} \vec{k}$
 13 $\vec{0}$
 15 $6\vec{i} + 4\vec{j} - 4\vec{k}$
 17 $-\pi\vec{i} - \pi\vec{k}$
 19 $9/\sqrt{3}$
 21 $-1/\sqrt{2}$
 23 $-\sqrt{77/2}$
 25 $-2\vec{i} - 2\vec{j} + 4\vec{k}$;
 $-2(x+1) - 2(y-1) + 4(z-2) = 0$
 27 $2\vec{j} - 4\vec{k}$; $2(y-1) - 4(z-2) = 0$
 29 $-2\vec{i} + \vec{k}$; $-2(x+1) + (z-2) = 0$
 31 (a) 0
 (b) $24/\sqrt{19}$
 33 (a) $(2x-yz)\vec{i} + (2y-xz)\vec{j} - xy\vec{k}$
 (b) $x+4y+6z=20$
 35 $z=12-4x-2y$
 37 $6(x-1) + 3(y-2) + 2(z-1) = 0$
 39 $2x+3y+2z=17$
 41 $z=2x+y+3$
 43 $x+4y+10z=18$
 45 (a) is (III); (b) is (I); (c) is (IV); (d) is (II)
 47 $22/\sqrt{14}$
 49 (a) Spheres centered at the origin
 (b) $2x \sin(x^2 + y^2 + z^2) \vec{i} + 2y \sin(x^2 + y^2 + z^2) \vec{j} + 2z \sin(x^2 + y^2 + z^2) \vec{k}$
 (c) $0, 180^\circ$
 51 (a) $-\vec{j}$,
 $\frac{2}{3}\vec{i} - \frac{1}{3}\vec{j} + \frac{2}{3}\vec{k}$
 (b) $y=0$; $2x-y+2z=3$
 (c) $x=y=0$ and $z \neq 0$
 53 (a) $1.2x + 1.6y + 6z = 8$
 (b) Yes; $\pm(0.8, 0.6, 1)$
 55 (a)



- (b) Valley
 57 (a) $-4\vec{i} + 4\vec{j} + 2\vec{k}$
 (b) $24 \text{ mg/cm}^3/\text{sec}$
 59 $f_{u_3} < f_{u_2} < 0 < f_{u_4} < f_{u_1}$
 61 (a) Spheres centered at origin
 (b) $(-2x\vec{i} - 2y\vec{j} - 2z\vec{k})e^{-(x^2+y^2+z^2)}$
 (c) $-3\sqrt{2}e^{-1}$ degrees/sec
 63 (a) is (V); (b) is (IV); (c) is (V)
 71 $f_x(0, 0, 0)x + f_y(0, 0, 0)y + f_z(0, 0, 0)z = 0$
 73 $f(x, y, z) = 2x + 3y + 4z + 100$
 75 False
 77 False
 79 (a) $^\circ\text{C}$ per meter
 (b) $^\circ\text{C}$ per second
 (c) $^\circ\text{C}$ per second

Section 14.6

$$1 \quad \frac{dz}{dt} = e^{-t} \sin(t)(2 \cos t - \sin t)$$

$$3 \quad 2 \cos \left(\frac{2t}{1-t^2} \right) \frac{1+t^2}{(1-t^2)^2}$$

$$5 \quad 2e^{1-t^2}(1-2t^2)$$

$$7 \quad \frac{\partial z}{\partial u} = \frac{1}{vu} \cos \left(\frac{\ln u}{v} \right) \\ \frac{\partial z}{\partial v} = -\frac{\ln u}{v^2} \cos \left(\frac{\ln u}{v} \right)$$

$$9 \quad \frac{\partial z}{\partial u} = \frac{e^v}{u} \\ \frac{\partial z}{\partial v} = e^v \ln u$$

$$11 \quad \frac{\partial z}{\partial u} = 2ue^{(u^2-v^2)}(1+u^2+v^2) \\ \frac{\partial z}{\partial v} = 2ve^{(u^2-v^2)}(1-u^2-v^2)$$

$$13 \quad \frac{\partial z}{\partial u} = (e^{-v} \cos u - v(\cos u)e^{-u \sin v}) \sin v \\ - (-u(\sin v)e^{-v \cos u} + e^{-u \sin v})v \sin u \\ \frac{\partial z}{\partial v} = (e^{-v} \cos u - v(\cos u)e^{-u \sin v})u \cos v \\ + (-u(\sin v)e^{-v \cos u} + e^{-u \sin v}) \cos u$$

$$15 \quad \frac{\partial z}{\partial u} = \frac{-2uv^2}{u^4 + v^4} \\ \frac{\partial z}{\partial v} = \frac{2vu^2}{u^4 + v^4}$$

$$17 \quad -2\rho \cos 2\phi, 0$$

$$19 \quad 401.1$$

$$21 \quad -5 \text{ pascal/hour}$$

$$23 \quad -0.6$$

$$25 \quad \text{Three}$$

$$27 \quad \frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt}$$

$$29 \quad b \cdot e + d \cdot p$$

$$35 \quad b \cdot e + d \cdot p$$

$$37 \quad (a) \quad \frac{\partial z}{\partial r} = \cos \theta \frac{\partial z}{\partial x} + \sin \theta \frac{\partial z}{\partial y} \\ \frac{\partial z}{\partial \theta} = r(\cos \theta \frac{\partial z}{\partial y} - \sin \theta \frac{\partial z}{\partial x}) \\ (b) \quad \frac{\partial z}{\partial y} = \sin \theta \frac{\partial z}{\partial r} + \frac{\cos \theta}{r} \frac{\partial z}{\partial \theta} \\ \frac{\partial z}{\partial x} = \cos \theta \frac{\partial z}{\partial r} - \frac{\sin \theta}{r} \frac{\partial z}{\partial \theta}$$

$$39 \quad \left(\frac{\partial U_3}{\partial P} \right)_V$$

$$41 \quad \left(\frac{\partial U}{\partial T} \right)_V = 7/2 \\ \left(\frac{\partial U}{\partial V} \right)_T = 11/4$$

$$45 \quad \int_0^b F_u(x, y) dy$$

$$49 \quad C_R \text{ and } C_T \text{ must be evaluated at } (5, 1)$$

$$51 \quad g(t) = 3 + t, h(t) = t$$

$$53 \quad z = x + y, x = e^t \text{ and } y = t^2, \text{ many other answers are possible}$$

$$55 \quad z = x + y, x = t \text{ and } y = t$$

$$5 \quad f_{xx} = 0 \\ f_{yx} = e^y = f_{xy} \\ f_{yy} = e^y(x+2+y)$$

$$7 \quad f_{xx} = -\left(\sin \left(\frac{x}{y}\right)\right) \left(\frac{1}{y^2}\right) \\ f_{xy} = -\left(\sin \left(\frac{x}{y}\right)\right) \left(\frac{-x}{y^2}\right) \left(\frac{1}{y}\right) \\ + \left(\cos \left(\frac{x}{y}\right)\right) \left(\frac{-1}{y^2}\right) = f_{yx} \\ f_{yy} = -\left(\sin \left(\frac{x}{y}\right)\right) \left(\frac{-x}{y^2}\right)^2 \\ + \left(\cos \left(\frac{x}{y}\right)\right) \left(\frac{2x}{y^3}\right)$$

$$9 \quad f_{xx} = 30xy^2 + 18 \\ f_{xy} = 30x^2y - 21y^2 \\ f_{yx} = 30x^2y - 21y^2 \\ f_{yy} = 10x^3 - 42xy$$

$$11 \quad f_{xx} = -12 \sin 2x \cos 5y \\ f_{xy} = -30 \cos 2x \sin 5y \\ f_{yx} = -30 \cos 2x \sin 5y \\ f_{yy} = -75 \sin 2x \cos 5y$$

$$13 \quad Q(x, y) = 1 + 2x - 2y + x^2 - 2xy + y^2$$

$$15 \quad Q(x, y) = 1 + x + x^2/2 - y^2/2$$

$$17 \quad Q(x, y) = 1 - x^2/2 - 3xy - (9/2)y^2$$

$$19 \quad Q(x, y) = -y + x^2 - y^2/2$$

$$21 \quad 1 + x - y/2 - x^2/2 + xy/2 - y^2/8$$

$$23 \quad (a) \quad \text{Negative}$$

$$(b) \quad \text{Zero}$$

$$(c) \quad \text{Negative}$$

$$(d) \quad \text{Zero}$$

$$(e) \quad \text{Zero}$$

$$25 \quad (a) \quad \text{Positive}$$

$$(b) \quad \text{Zero}$$

$$(c) \quad \text{Positive}$$

$$(d) \quad \text{Zero}$$

$$(e) \quad \text{Zero}$$

$$27 \quad (a) \quad \text{Zero}$$

$$(b) \quad \text{Negative}$$

$$(c) \quad \text{Zero}$$

$$(d) \quad \text{Negative}$$

$$(e) \quad \text{Zero}$$

$$29 \quad (a) \quad \text{Positive}$$

$$(b) \quad \text{Positive}$$

$$(c) \quad \text{Zero}$$

$$(d) \quad \text{Zero}$$

$$(e) \quad \text{Zero}$$

$$31 \quad (a) \quad \text{Positive}$$

$$(b) \quad \text{Negative}$$

$$(c) \quad \text{Negative}$$

$$(d) \quad \text{Negative}$$

$$(e) \quad \text{Positive}$$

$$33 \quad L(x, y) = y$$

$$Q(x, y) = y + 2(x-1)y$$

$$L(0.9, 0.2) = 0.2$$

$$Q(0.9, 0.2) = 0.16$$

$$f(0.9, 0.2) = 0.162$$

$$39 \quad a = -b^2$$

$$41 \quad \text{Positive, negative}$$

$$43 \quad (a) \quad z_{yx} = 4y$$

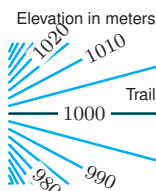
$$(b) \quad z_{xyx} = 0$$

$$(c) \quad z_{xyy} = 4$$

$$45 \quad d = e = f = 0$$

$$47 \quad d = 0, e > 0, f < 0$$

$$49 \quad (a)$$



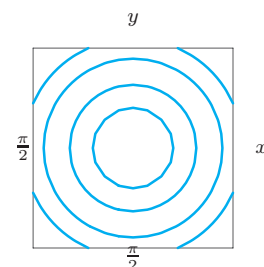
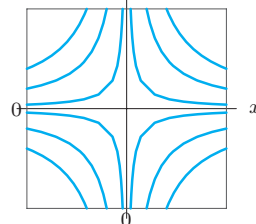
$$(b) \quad \frac{\partial h}{\partial x} = 0, \quad \frac{\partial h}{\partial y} > 0, \\ (\partial^2 h)/(\partial x \partial y) < 0$$

$$(c) \quad (\partial^2 h)/(\partial x \partial y)$$

$$51 \quad (a) \quad A$$

$$(b) \quad B$$

$$53 \quad (a) \quad xy \\ 1 - \frac{1}{2}(x - \frac{\pi}{2})^2 - \frac{1}{2}(y - \frac{\pi}{2})^2 \\ (b)$$



$$55 \quad (a) \quad (i) \quad \text{Dollars/Year, negative,} \quad (ii) \quad \text{Dollars/Dollar, positive}$$

$$(b) \quad \partial^2 P/(\partial A \partial C) < 0$$

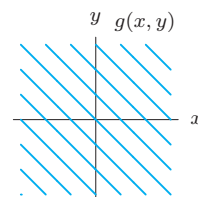
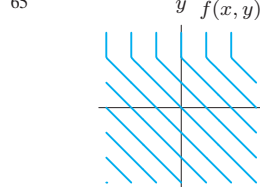
$$(c) \quad eCA$$

$$57 \quad (a) \quad 1 + x/2 + y$$

$$(b) \quad 1 + x/2 + y - x^2/8 - xy/2 - y^2/2$$

$$61 \quad \text{Counterexample: } f(x, y) = x^3 + y^4$$

$$63 \quad f(x, y) = x^2 + y^2$$



Section 14.7

$$1 \quad f_{xx} = 2$$

$$f_{yy} = 2$$

$$f_{yx} = 2$$

$$f_{xy} = 2$$

$$3 \quad f_{xx} = 6y$$

$$f_{xy} = 6x + 15y^2$$

$$f_{yx} = 6x + 15y^2$$

$$f_{yy} = 30xy$$

Section 14.8

$$1 \quad (0, 0)$$

$$3 \quad x\text{-axis and } y\text{-axis}$$

$$5 \quad \text{None}$$